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Properties of $\hat{\beta}$

↳ Gauss-Markov Thm.

→ efficiency of $\hat{\beta}$ in which homoskedasticity holds

→ given linear regression assumptions and homoskedasticity, for any possible linear unbiased estimator of β ($\tilde{\beta}$), $\text{var}(\tilde{\beta} | X) \geq \text{var}(\hat{\beta} | X)$

① Linearity - $\tilde{\beta} = \underline{A}^T \underline{Y}$ where $\underline{A}$ is an $n \times k$ matrix that's a function of $\underline{X}$. $\hat{\beta}$ has $\underline{A}^T = (\underline{X}^T \underline{X})^{-1} \underline{X}^T$.

② Unbiasedness - $E[\tilde{\beta} | \underline{X}] = \beta$

$$\begin{aligned} 1+2: E[\underline{A}^T \underline{Y} | \underline{X}] &\Rightarrow \underline{A}^T E[\underline{Y} | \underline{X}] = \underline{A}^T \underline{X} \beta = \beta \\ &\Rightarrow \underline{A}^T \underline{X} = \underline{I}_k \end{aligned}$$

$$\begin{aligned} \text{var}(\tilde{\beta} | \underline{X}) &= \text{var}(\underline{A}^T \underline{Y} | \underline{X}) = \text{var}(\underline{A}^T (\underline{X} \beta + \underline{e}) | \underline{X}) \\ &= \text{var}(\underline{A}^T \underline{e} | \underline{X}) = \underline{A}^T \text{var}(\underline{e} | \underline{X}) \underline{A} \\ &= \sigma^2 \underline{A}^T \underline{A} \end{aligned}$$

now show $\underline{A}^T \underline{A} - (\underline{X}^T \underline{X})^{-1} \geq 0$

$$\star = \underline{A}^T \underline{A} - \underline{A}^T \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{A}$$

$$= \underline{A}^T (\underline{I} - \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T) \underline{A}$$

$$= \underline{A}^T \underline{M} \underline{A} = \underline{A}^T \underline{M}^T \underline{M} \underline{A}$$

$$= (\underline{M} \underline{A})^T \underline{M} \underline{A} \geq 0$$

↳ akin to quadratic form

GLS:

$$\hookrightarrow \underline{D} = \text{var}(\underline{e} | \underline{X})$$

$\sigma^2 \underline{\Sigma}$ (allows off-diagonal to be $\neq 0$)
↗ scalar

$$\text{var}(\hat{\beta} | \underline{X}) = \sigma^2 (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{\Sigma} \underline{X} (\underline{X}^T \underline{X})^{-1}$$

↳ Thm. 4.5: Prove on hw

→ show for any linear unbiased est of β ($\tilde{\beta}$), $\text{var}(\tilde{\beta} | \underline{X}) \geq \sigma^2 (\underline{X}^T \underline{\Sigma} \underline{X})^{-1}$

↳ Suppose $\underline{\Sigma}$ is known:

$$\rightarrow \tilde{Y} = \underline{\Sigma}^{-\frac{1}{2}} \underline{Y}$$

$$\rightarrow \tilde{X} = \underline{\Sigma}^{-\frac{1}{2}} \underline{X}$$

$$\rightarrow \tilde{\beta}_{GLS} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{Y}$$

GLS:

$$\tilde{\beta} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \tilde{Y}$$

$$= ((\Sigma^{-\frac{1}{2}} \tilde{X})^T \Sigma^{-\frac{1}{2}} \tilde{X}) (\Sigma^{-\frac{1}{2}} \tilde{X})^T \Sigma^{-\frac{1}{2}} \tilde{Y}$$

$$= (\tilde{X}^T \Sigma^{-1} \tilde{X})^{-1} \tilde{X}^T \Sigma^{-1} \tilde{Y}$$

$$\text{var}(\tilde{\beta}_{\text{GLS}} | X) = \sigma^2 (X^T \Sigma^{-1} X)^{-1}$$

Feasible GLS:

↳ infrequently used due to instability

↳ use $\hat{\Sigma}$ as estimator of Σ

Asymptotic Theory for Least Squares

Def: An estimator $\hat{\theta}$ of θ is consistent if $\hat{\theta} \xrightarrow{P} \theta$

↳ use WLLN to show consistency

→ if $Y_i \in \mathbb{R}^k$ are i.i.d and $E \|Y\| < \infty$

$$\text{then } \frac{1}{n} \sum_{i=1}^n Y_i \xrightarrow{P} E[Y]$$

Def: An estimator $\hat{\theta}$ of θ is asymptotically normal if $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \underline{V})$ for some $\underline{V}$

$\hookrightarrow$ Central Limit Thm.: if $Y_i \in \mathbb{R}^k$ are i.i.d., $E\|Y\|^2 < \infty$, $\sqrt{n}(\frac{1}{n} \sum_{i=1}^n Y_i - E[Y]) \xrightarrow{d} N(0, \underline{V})$ s.t.
 $\underline{V} = \text{var}(Y)$